ASSIGNMENT No. 03

Conditional Statements: *Develop menu-driven C programs using if, if-else, nested if, and switch-case to solve problems such as grading, electricity bill calculation, simple calculator, and leap year identification.*

1. For any character entered through keyboard check whether the entered character is capital letter, small letter, a digit or special symbol.
2. To find the greatest of 3 integers using two methods (a) if-else and (b) ternary operator.
3. To implement calculator (+, -/, *, %) using switch statements.
4. Display month name according to the month number using Switch Statement.
5. To check whether a number is: (a) even or odd (b) negative or positive by using (i) if-else (ii) ternary operator.
6. To check a no. say 'n' divisible by 5 and/or 8. Print the appropriate message accordingly.
7. A library charges a fine for every book returned late. For the first 5 days the fine is 50 paisa, for 6-10 days fine is one rupee and above 10 days fine is 5 rupees. If you return the book after 30 days your membership will be canceled. WAP to accept no. of days the member is late to return the book and display the fine or appropriate message.
8. Write a program to read in 5 numbers and compute the average, max, min, standard deviation.
9. Using if-else decision making, write a program to read number of computers that need services from the user and calculate the service charge as per the information shown below.

PCs	Charges / per pc
Up to 3	100
4-6	75
7-9	50
>10	40

10. Using if-else decision making, write a C program that reads an integer number (1–12) representing a month of the year and displays the total number of days in that month. If the user enters a number outside the range of 1 to 12, display an appropriate error message.
11. Using if-else decision making, write a program in C that displays the recommended actions depending on the color of a traffic light.

Color	Recommended Actions
Red ('R' or 'r')	STOP
Yellow ('Y' or 'y')	CAUTION
Green ('G' or 'g')	GO
Default	Wrong Color Selected

12. A bank provides loans at a base interest rate of 15%. The bank also offers rebates under the following conditions:
    - 2% rebate if the applicant is from a rural area.
    - 2% rebate if the applicant belongs to the below-poverty-line (BPL) category.
    - 2% rebate if the loan is taken for agriculture, fishery, or dairy purposes.

    Write a C program that reads the details of a person (whether they are from a rural area, belong to BPL, and the purpose of the loan) and computes the final interest rate applicable to the loan.

13. Write a C program using a switch statement to print the lowercase English word corresponding to a number entered by the user. If the input number n is between 1 and 9 (inclusive), print its English word in lowercase (e.g., “one” for 1, “two” for 2, etc.). For any number greater than 9, print “greater than 9”.